

Samart's conjecture $n_4(81) = 40M_7$: the exact CM evaluation and the two obstructions

A status report, companion to *Mahler measures at interior CM points*

Huimin Zheng*

Declaration on the use of AI tools

The research reported in this note — including the computational exploration, the discovery of the proof strategy, the machine-certified verifications, and the preparation of the manuscript — was carried out by the author with the assistance of the AI system *Kimi* (Moonshot AI). All mathematical content, including every proof and every certified computation, has been checked and verified by the author, who takes full responsibility for the correctness and integrity of the note. All verification scripts cited below are available for independent verification.

Abstract

This note archives the status of Samart's Table-6 conjecture

$$n_4(81) \stackrel{?}{=} 40M_7, \quad M_7 := L'(g_7, 0),$$

where $g_7(\tau) = \eta(\tau)^3 \eta(7\tau)^3$ is the newform of $S_3(\Gamma_0(7), \chi_{-7})$ and $n_4(s) := 4m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz)$. The conjecture is the cleanest of Samart's open interior-point entries (discriminant -7 , class number 1, a *single* L -value), and it was dropped as a theorem target in the companion paper [1], where the two n_2 -family conjectures were proved. We record what is proved and precisely where the methods of [1] fail. First, a complete proof of the L -value side (P1): at the CM point $\tau_2 = (7 + \sqrt{-7})/4$ the Eisenstein–Kronecker expression underlying Samart's formula evaluates *exactly* to

$$\text{EK}_4(\tau_2) = 40M_7,$$

via lattice sums over the ring of integers of $\mathbb{Q}(\sqrt{-7})$ and the principal ideal $(\bar{\pi})$, with an exact cancellation of the parasitic $\zeta_K(2)$ -terms. Second, two quantitative obstructions to the remaining half $n_4(81) = \text{EK}_4(\tau_2)$: the critical image of the n_4 -family is a two-dimensional astroid disc containing the parameter $c = 3$ in its interior (in contrast to the one-dimensional slit $[0, 64]$ of [1]), so no continuation path can approach the CM point; and Samart's U -series converges on all of $\mathbb{H}$ but leaves the geometric sheet of the holomorphic Mahler measure everywhere below $\text{Im } \tau = 1/\sqrt{2}$, so the premise of the differential-comparison continuation fails. A 20-digit direct torus integration then decides the conjecture numerically: $n_4(81) - 40M_7 = +0.0586706795972872\dots$, five orders of magnitude above the integration error floor, so the identity as literally stated is refuted; a closed form for the true value $n_4(81)$ remains open and appears to require regulator/monodromy machinery.

*College of Information and Network Engineering, Anhui Science and Technology University, Fengyang, Anhui 233100, P. R. China. Email: zhhm@ahstu.edu.cn.

1 Introduction

1.1 The conjecture

Let

$$P(x, y, z) := \frac{x^4 + y^4 + z^4 + 1}{xyz},$$

and denote by m the logarithmic Mahler measure,

$$m(Q) = \int_{\mathbb{T}^3} \log |Q(x, y, z)| \frac{dx}{2\pi i x} \frac{dy}{2\pi i y} \frac{dz}{2\pi i z}, \quad \mathbb{T} = \{|w| = 1\}.$$

Since $m(xyz) = 0$, one has $m(P + c) = m(x^4 + y^4 + z^4 + 1 + cxyz)$, and the substitution $x \mapsto ix$ (Haar-invariant, sending $c \mapsto ic$) shows that the value is unchanged when c is multiplied by a fourth root of unity. Following Samart's table notation [3] we therefore write, unambiguously,

$$n_4(s) := 4m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz), \quad s \in \mathbb{C}, \quad (1)$$

for either choice of the fourth root. (In [2] the same function is denoted f_4 .)

Conjecture (Samart, [3, Table 6]). $n_4(81) = 40M_7$, where $M_7 := L'(g_7, 0)$ and $g_7(\tau) = \eta(\tau)^3 \eta(7\tau)^3$ is the unique normalized newform of $S_3(\Gamma_0(7), \chi_{-7})$ (LMFDB label 7.3.b.a [10]).

Among Samart's open interior-point entries this one is structurally the cleanest: the attached CM point has discriminant -7 (class number 1), and the conjectural value is a *single* L -value with no Dirichlet term, in contrast to the two-term formula of the n_2 conjugate pair proved in [1]. Fei [4] proved the identity at the level of the real part of the holomorphic (modified) Mahler measure $\operatorname{Re} \tilde{m}$; explicitly, in Fei's table the row $c = 81$ is placed between a pair of double lines marking the entries for which $\operatorname{Re} \tilde{m}$ is *not* guaranteed to equal the genuine Mahler measure [4, Note (3) and Caution 2.3]. The passage to the genuine Mahler measure at the interior parameter $s^{1/4} = 3$ was left open, and Section 6 below determines what actually happens there: the two sides differ in the fifth significant digit.

1.2 What this note records

The companion paper [1] proved Samart's conjectures for the n_2 -family at $s = 1$ (Theorem A, differential-comparison continuation) and at $s = (47 \pm 45\sqrt{-7})/2$ (Theorem B, the Fricke trick). The present conjecture was considered as a third target and *dropped*: two independent obstructions, documented quantitatively in Sections 4 and 5 below, kill both the continuation machine and the Fricke shortcut for the n_4 -family. This note archives the state of the problem for future work:

- (P1) **Done, exact.** The L -value side. Samart's Eisenstein–Kronecker expression EK_4 at the CM point $\tau_2 = (7 + \sqrt{-7})/4$ (where $s_4(\tau_2) = 81$) evaluates exactly to $40M_7$ (Theorem 3.4, full proof in Section 3; certified to 2.7×10^{-59} by the 45 checks of `verify_P1_n4_81.py`).
- (P2) **Blocked.** The Mahler side $n_4(81) \stackrel{?}{=} \operatorname{EK}_4(\tau_2)$. Obstruction 1: the critical image $P(\mathbb{T}^3)$ is the two-dimensional *astroid disc* $\{| \operatorname{Re} c |^{2/3} + | \operatorname{Im} c |^{2/3} \leq 4^{2/3}\}$, and $c = 3$ is an interior point, so the open mapping theorem forbids any path continuation (Section 4). Obstruction 2, the decisive one: Samart's U -series is proved to equal $n_4(s_4(\tau))$ only for $\operatorname{Im} \tau \geq 1/\sqrt{2}$, and below that threshold it leaves the geometric sheet of the holomorphic Mahler measure *everywhere* — not only inside the critical image (Section 5).

In one sentence: for the n_2 -family the bad locus was the one-dimensional slit $[0, 64]$, which a certified path can go around; for the n_4 -family the bad locus is a two-dimensional disc *and* the analytic formula itself is on the wrong sheet below $\text{Im } \tau = 1/\sqrt{2}$, so there is nothing left to continue.

1.3 Notation

We keep the notation of [1]. Throughout, $\tau = x + iy \in \mathbb{H}$, $q = e^{2\pi i \tau}$, $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$, and $\sum'$ denotes omission of the zero term. The field $K = \mathbb{Q}(\sqrt{-7})$ has ring of integers $\mathcal{O}_K = \mathbb{Z}[\pi]$, class number $h(-7) = 1$ and units $\{\pm 1\}$, where

$$\begin{aligned} \pi &:= \frac{1 + \sqrt{-7}}{2}, & \pi + \bar{\pi} &= 1, & \pi \bar{\pi} &= N(\pi) = 2, & \pi^2 &= \pi - 2, \\ \pi^2 + \bar{\pi}^2 &= -3, & \text{Re } \pi^2 &= \text{Re } \bar{\pi}^2 = -\frac{3}{2}, & (\bar{\pi}) &= \bar{\pi} \mathcal{O}_K = 2\mathbb{Z} + \bar{\pi}\mathbb{Z}. \end{aligned} \quad (2)$$

We write $L_3 := L(g_7, 3)$. The theta identity [1, Appendix B] gives

$$\sum'_{\alpha \in \mathcal{O}_K} \frac{\alpha^2}{|\alpha|^{2s}} = 2L(g_7, s), \quad \sum'_{\alpha \in \mathcal{O}_K} \frac{\bar{\alpha}^2}{|\alpha|^6} = 2L_3, \quad \sum'_{\alpha \in \mathcal{O}_K} |\alpha|^{-4} = 2\zeta_K(2), \quad (3)$$

the middle equality by conjugation ($L_3 \in \mathbb{R}$ since $a_n(g_7) \in \mathbb{Z}$), and the functional equation of g_7 (root number +1) gives

$$M_7 = L'(g_7, 0) = \frac{7\sqrt{7}}{4\pi^3} L_3. \quad (4)$$

Both (3) and (4) are quoted theorems, certified numerically in [1] (`verify_P1.py`, `lvalue_g7.py`) and re-certified for this note in `verify_P1_n4_81.py` (check [L1]: the theta identity to q^{60} as exact integers, (4) to 10^{-61}).

2 The EK_4 expression and Samart's theorem

2.1 Two forms of EK_4

For $z \in \mathbb{C} \setminus \{0\}$ let

$$F(z) := \frac{4(\text{Re } z)^2}{|z|^6} - \frac{1}{|z|^4},$$

and for $\tau \in \mathbb{H}$ and $d \geq 1$ define the double series

$$T_d(\tau) := \sum'_{(m,n) \in \mathbb{Z}^2} F(dm\tau + n), \quad (5)$$

summed by rows in m ; each row is absolutely convergent and the row sums decay like $|m|^{-2}$ (Poisson summation), and the $m = 0$ row is $\sum_{n \neq 0} (4n^2 \cdot n^{-6} - n^{-4}) = 6\zeta(4) = \pi^4/15$ [1, §2.4]. Define, for all $\tau \in \mathbb{H}$,

$$\text{EK}_4(\tau) := \frac{10 \text{Im } \tau}{\pi^3} (-T_1(\tau) + 4T_2(\tau)), \quad (6)$$

real-analytic on all of $\mathbb{H}$. Since $4(\text{Re } \lambda)^2 = (\lambda + \bar{\lambda})^2 = \lambda^2 + 2|\lambda|^2 + \bar{\lambda}^2$, the summand rewrites as $F(\lambda) = 2 \text{Re}(\bar{\lambda}^2/|\lambda|^6) + |\lambda|^{-4}$, so with $\Lambda_d(\tau) := \mathbb{Z} + \mathbb{Z} d\tau$,

$$T_d(\tau) = \sum'_{\lambda \in \Lambda_d(\tau)} \left[2 \text{Re} \frac{\bar{\lambda}^2}{|\lambda|^6} + \frac{1}{|\lambda|^4} \right], \quad (7)$$

the form used for the CM evaluation.

Define also, for $j \geq 1$,

$$U_j(\tau) := \sum_{m \neq 0} \frac{1}{m} \sum_{n \in \mathbb{Z}} (jm\tau + n)^{-3} = 2\pi^3 \sum_{m \geq 1} \frac{\cos(j\pi m\tau)}{m \sin^3(j\pi m\tau)},$$

absolutely convergent on $\mathbb{H}$ exactly as in [1, §2.4].

Lemma 2.1 (The two forms agree). *For every $\tau \in \mathbb{H}$,*

$$\text{EK}_4(\tau) = \text{Im} \left[2\pi\tau + \frac{10}{\pi^3} (U_1(\tau) - 2U_2(\tau)) \right]. \quad (8)$$

Proof. The computation of [1, Lemma (key identity)] gives $\text{Im } U_j = -jy(T_j(\tau) - \pi^4/15)$, $y = \text{Im } \tau$. Substituting,

$$\text{Im} \left[2\pi\tau + \frac{10}{\pi^3} (U_1 - 2U_2) \right] = 2\pi y + \frac{10}{\pi^3} \left[-y(T_1 - \frac{\pi^4}{15}) + 4y(T_2 - \frac{\pi^4}{15}) \right],$$

and the $m = 0$ -row contribution is $\frac{10y}{\pi^3} \cdot \frac{\pi^4}{15} \cdot (1 - 4) = -2\pi y$, cancelling the polynomial term $2\pi y$ and leaving (6). (Numerically the two forms agree at τ_2 , i, $i/\sqrt{2}$ to 10^{-58} ; `verify_P1_n4_81.py`, checks [EK], [A2], [A4].) $\square$

The U -series form (8) is Samart's exponential-sum formula for the n_4 -family; it converges for every $\tau \in \mathbb{H}$.

2.2 The parametrization and Samart's theorem

Define

$$c(\tau) := \left(\frac{\eta(2\tau)}{\eta(\tau)} \right)^6 (16A(\tau)^4 + A(\tau)^{-4}), \quad A(\tau) := \frac{\eta(\tau)\eta(4\tau)^2}{\eta(2\tau)^3}, \quad (9)$$

holomorphic and nonvanishing-free on $\mathbb{H}$, and $s_4(\tau) := c(\tau)^4$ (this agrees with Samart's Δ -quotient definition of his signature-4 modular function, checked numerically to 40 digits at sample points; `diag_n4_astroid.py`, Part 1). With Samart's normalization [2, Lemma 2.2],

$$s_4(i) = 648, \quad s_4(i/\sqrt{2}) = 256, \quad s_4(iy) \geq 256 \quad (y > 0),$$

so on the imaginary axis $c(iy)$ is real with $c(iy) \geq 4$.

Theorem 2.2 (Samart, [2, Prop. 2.1(iii)]). *For $\tau \in \mathbb{H}$ with $\text{Im } \tau \geq 1/\sqrt{2}$,*

$$n_4(s_4(\tau)) = \text{EK}_4(\tau). \quad (10)$$

We take this as a quoted theorem. Its proof region is essential to what follows: the identity is *not* asserted, and in fact fails, below $\text{Im } \tau = 1/\sqrt{2}$ (Section 5).

Remark 2.3 (Anchors). The EK_4 machinery is anchored against *proved* values. At $\tau = i$ ($\text{Im } \tau = 1 \geq 1/\sqrt{2}$), (10) applies and Samart's Theorem 1.4 (1.9) of [2] gives

$$\text{EK}_4(i) = n_4(648) = \frac{160}{\pi^3} L(h, 3) + \frac{5}{\pi} L(\chi_{-4}, 2), \quad h(\tau) := \eta(4\tau)^6 \in S_3(\Gamma_0(16), \chi_{-4}), \quad (11)$$

confirmed five independent ways (the lattice form (6), the U -series (8), the L -value formula (11), Rogers' hypergeometric evaluation of $n_4(648)$ [2, 5], and direct torus integration of $4m(P + 648^{1/4})$, the last at $\tanh$ -sinh degree 5 to 2×10^{-21}), all agreeing to 10^{-58} : $\text{EK}_4(i) = 4m(P + 648^{1/4}) = 6.4332830658\dots$. On the boundary, $\text{EK}_4(i/\sqrt{2}) = n_4(256)$ via Rogers' ${}_5F_4$ at $|k| = 256$ (convergent since $\sum b - \sum a = 3/2 > 0$). `verify_P1_n4_81.py`, checks [A0]–[A4].

3 (P1) The exact CM evaluation at τ_2

Throughout this section

$$\tau_2 := \frac{7 + \sqrt{-7}}{4}, \quad \text{Im } \tau_2 = \frac{\sqrt{7}}{4}, \quad \tau_w := \frac{-1 + \sqrt{-7}}{4}$$

(τ_w is the Fricke CM point of Theorem B in [1]). Then

$$2\tau_2 = \frac{7 + \sqrt{-7}}{2} = 3 + \pi, \quad \tau_2 = \tau_w + 2. \quad (12)$$

Numerically $s_4(\tau_2) = 81$ to 10^{-58} (`verify_P1_n4_81.py`, check [A0]); its exactness — a class invariant computation in the spirit of [1, §4.2] — is outside the scope of this note and is *not* used below: (P1) is the exact evaluation of the concrete series $\text{EK}_4(\tau_2)$, independently of the value of s_4 .

3.1 The lattices

Lemma 3.1. $\Lambda_2(\tau_2) = \mathcal{O}_K$ and $\Lambda_1(\tau_2) = (\bar{\pi})/2$, where $(\bar{\pi}) = \bar{\pi}\mathcal{O}_K$ is the principal ideal of norm 2.

Proof. By (12), $\Lambda_2(\tau_2) = \mathbb{Z} + \mathbb{Z}2\tau_2 = \mathbb{Z} + \mathbb{Z}(3 + \pi) = \mathbb{Z} + \mathbb{Z}\pi = \mathcal{O}_K$. For the second, an integer translate of τ does not change the lattice $\Lambda_1(\tau) = \{m\tau + n\}$, so $\Lambda_1(\tau_2) = \Lambda_1(\tau_w)$. Since $2\tau_w = (-1 + \sqrt{-7})/2 = -\bar{\pi}$,

$$2\Lambda_1(\tau_w) = 2\mathbb{Z} + \mathbb{Z}2\tau_w = 2\mathbb{Z} + \bar{\pi}\mathbb{Z} = (\bar{\pi})$$

by (2). Hence $\Lambda_1(\tau_2) = (\bar{\pi})/2$, homothetic to the principal ideal $(\bar{\pi})$ of norm 2. (Note the conjugation: identifying $2\Lambda_1$ directly from $2\tau_2 = 3 + \pi$ would give $\mathbb{Z} + \pi\mathbb{Z} = \mathcal{O}_K$, since $\{3m + 2n : m, n \in \mathbb{Z}\} = \mathbb{Z}$ — the route through τ_w is what pins the ideal $(\bar{\pi})$. The T -values are unaffected since $\text{Re } \bar{\pi}^2 = \text{Re } \pi^2$.) $\square$

Both lattices are $\mathcal{O}_K$ -ideals: unlike the τ_w evaluation of [1, §5] there is no imprimitive order and no Euler factor at the primes above 2.

3.2 The Hecke sums

For a fractional $\mathcal{O}_K$ -ideal $\mathfrak{a}$ define

$$B(\mathfrak{a}) := \sum'_{\gamma \in \mathfrak{a}} |\gamma|^{-4}, \quad G(\mathfrak{a}) := \sum'_{\gamma \in \mathfrak{a}} \frac{\bar{\gamma}^2}{|\gamma|^6}.$$

By (3),

$$B(\mathcal{O}_K) = 2\zeta_K(2), \quad G(\mathcal{O}_K) = 2L_3. \quad (13)$$

For the principal ideal $(\bar{\pi})$ write $\alpha = \bar{\pi}\gamma$, $\gamma \in \mathcal{O}_K \setminus \{0\}$; then $|\alpha|^{-4} = |\bar{\pi}|^{-4}|\gamma|^{-4}$ and $\bar{\alpha}^2|\alpha|^{-6} = (\pi^2/|\bar{\pi}|^6)\bar{\gamma}^2|\gamma|^{-6}$, so with $N(\bar{\pi}) = 2$ and $|\bar{\pi}|^6 = 8$,

$$B((\bar{\pi})) = \frac{2\zeta_K(2)}{N(\bar{\pi})^2} = \frac{\zeta_K(2)}{2}, \quad G((\bar{\pi})) = \frac{\pi^2}{8} \cdot 2L_3 = \frac{\pi^2 L_3}{4}. \quad (14)$$

Note $G((\bar{\pi}))$ is genuinely complex ($(\bar{\pi})$ is not conjugation-stable); only its real part enters the T -sums, and by (2),

$$\text{Re } G((\bar{\pi})) = \frac{\text{Re } \pi^2}{4} L_3 = -\frac{3}{8} L_3. \quad (15)$$

(Direct Poisson row summation confirms (13)–(15), including the imaginary part $\text{Im } G((\bar{\pi})) = \sqrt{7}L_3/8$, to $\sim 10^{-60}$; `verify_P1_n4_81.py`, checks [S1]–[S4].)

3.3 The T -sums and the combination

Proposition 3.2. *With $L_3 := L(g_7, 3)$,*

$$T_2(\tau_2) = 4L_3 + 2\zeta_K(2), \quad T_1(\tau_2) = -12L_3 + 8\zeta_K(2). \quad (16)$$

Proof. By (7) with $\Lambda_2(\tau_2) = \mathcal{O}_K$ and (13),

$$T_2(\tau_2) = 2 \operatorname{Re} G(\mathcal{O}_K) + B(\mathcal{O}_K) = 4L_3 + 2\zeta_K(2).$$

For $\Lambda_1(\tau_2) = (\bar{\pi})/2$ rescale $\lambda = \alpha/2$, $\alpha \in (\bar{\pi}) \setminus \{0\}$: then $|\lambda|^{-4} = 16|\alpha|^{-4}$ and $\bar{\lambda}^2|\lambda|^{-6} = 16\bar{\alpha}^2|\alpha|^{-6}$, so by (14) and (15),

$$T_1(\tau_2) = 16 \left[2 \operatorname{Re} G((\bar{\pi})) + B((\bar{\pi})) \right] = 16 \left[-\frac{3L_3}{4} + \frac{\zeta_K(2)}{2} \right] = -12L_3 + 8\zeta_K(2). \quad \square$$

Corollary 3.3 (Exact cancellation of $\zeta_K(2)$).

$$-T_1(\tau_2) + 4T_2(\tau_2) = (12 + 16)L_3 + (-8 + 8)\zeta_K(2) = 28L_3. \quad (17)$$

Unlike the τ_w case of [1] (where the surviving $\zeta_K(2)$ -term produced the Dirichlet value d_7), the coefficients -8 and $+8$ cancel exactly, leaving a *single* L -value — the structural prerequisite for the one-term formula $40M_7$.

Theorem 3.4 ((P1) at τ_2).

$$\operatorname{EK}_4(\tau_2) = 40M_7.$$

Proof. With $\operatorname{Im} \tau_2 = \sqrt{7}/4$, (6) and (17),

$$\operatorname{EK}_4(\tau_2) = \frac{10 \cdot \sqrt{7}/4}{\pi^3} \cdot 28L_3 = \frac{70\sqrt{7}}{\pi^3} L_3 = \frac{70\sqrt{7}}{\pi^3} \cdot \frac{4\pi^3}{7\sqrt{7}} M_7 = 40M_7,$$

the last equality being (4). The coefficient identity is $70 = 40 \cdot 7/4$; no transcendental input beyond the quoted functional equation (4) enters. $\square$

Remark 3.5 (Certification and cross-links). All steps are re-verified by `verify_P1_n4_81.py` (60 dps, 45 checks, all pass, worst difference 2.7×10^{-59}): the Hecke sums by direct row summation, the T -sums by direct Poisson-row evaluation of the defining double series (5) (independent of the ideal decomposition), and Theorem 3.4 via both forms (6) and (8). Two cross-links with [1]: EK_4 is 1-periodic (the U_j are, and $\operatorname{Im}[2\pi(\tau + 1)] = \operatorname{Im}[2\pi\tau]$), so $\operatorname{EK}_4(\tau_2) = \operatorname{EK}_4(\tau_w)$; and $T_1(\tau_2) = T_1(\tau_w) = -12L_3 + 8\zeta_K(2)$ is *numerically the same sum* as the T_1 of Theorem B, there evaluated over the same lattice $(\bar{\pi})/2$.

4 Obstruction 1: the critical image is two-dimensional

For the n_2 -family of [1] the bad locus of the parameter was the segment $[-8, 8]$ in the shift variable (equivalently the slit $[0, 64]$ in Samart's variable) — one-dimensional, with complement in $\mathbb{C}$ connected, so a certified path could thread around it. For the n_4 -family the situation is qualitatively different.

Write $x = e^{it_1}$, $y = e^{it_2}$, $z = e^{it_3}$ on $\mathbb{T}^3$. Then

$$P(x, y, z) = \frac{x^4 + y^4 + z^4 + 1}{xyz} = u_1 + u_2 + u_3 + u_4, \quad u_1 = \frac{x^4}{xyz}, \dots, u_4 = \frac{1}{xyz}, \quad (18)$$

where $|u_j| = 1$ and $u_1 u_2 u_3 u_4 = 1$; conversely every quadruple of unit complex numbers with product 1 occurs. Hence $P(\mathbb{T}^3)$ is the image of the four-phaser map.

Proposition 4.1 (The critical image; recorded computation). $P(\mathbb{T}^3)$ is the closed astroid disc

$$\mathcal{D} := \{c \in \mathbb{C} : |\operatorname{Re} c|^{2/3} + |\operatorname{Im} c|^{2/3} \leq 4^{2/3}\}, \quad (19)$$

whose boundary is the astroid

$$t \mapsto 3e^{it} + e^{-3it} = (4\cos^3 t, 4\sin^3 t), \quad (20)$$

with cusps at ± 4 and $\pm 4i$.

Justification. The boundary curve is exact algebra: the admissible family $u_1 = u_2 = u_3 = e^{it}$, $u_4 = e^{-3it}$ sums to $3e^{it} + e^{-3it}$, and $3\cos t + \cos 3t = 4\cos^3 t$, $3\sin t - \sin 3t = 4\sin^3 t$. That the full image is exactly the enclosed disc is a recorded computational fact of the project archive (`diag_n4_astroid.py`): dense sampling of admissible quadruples (2×10^5 samples) gives $\max(|\operatorname{Re} c|^{2/3} + |\operatorname{Im} c|^{2/3}) = 2.5198357\dots$ against $4^{2/3} = 2.5198421\dots$, with the bound approached along the curve (20), and the image filling the disc. We use the statement only as the geometric description of the critical locus; no proof step of this note depends on its sharpness. $\square$

The parameter of the conjecture sits strictly inside:

$$c = 3 : \quad |3|^{2/3} = 2.08008\dots < 4^{2/3} = 2.51984\dots, \quad (21)$$

i.e. $3 \in \operatorname{int} \mathcal{D}$; the zeros of $x^4 + y^4 + z^4 + 1 + 3xyz$ on $\mathbb{T}^3$ form a nonempty real-analytic set (the integrand of the direct Mahler integral is log-singular there; see Section 6).

Proposition 4.2 (No path continuation is possible). *Let $c(\tau)$ be the branch (9), with $c(\tau_2) = 3$ (from $s_4(\tau_2) = 81$, §3). Then τ_2 lies in the interior of the bad preimage $c^{-1}(\operatorname{int} \mathcal{D})$; in particular τ_2 is not in the closure of the good region $\mathcal{G} := \{\tau \in \mathbb{H} : c(\tau) \notin \mathcal{D}\}$, and no continuous path can approach τ_2 through $\mathcal{G}$.*

Proof. c is holomorphic and non-constant near τ_2 . By the open mapping theorem, c maps every sufficiently small neighborhood of τ_2 onto a neighborhood of $c(\tau_2) = 3$, and since $3 \in \operatorname{int} \mathcal{D}$ that neighborhood lies inside $\operatorname{int} \mathcal{D}$. Hence a full neighborhood of τ_2 lies in the bad preimage. $\square$

Remark 4.3 (Contrast with the n_2 -family). In [1] the certified-path criterion required exactly the membership of the target CM point in the closure of the good component W , supplied there by an explicit 99-block interval certificate — possible because the bad slit $[0, 64]$ has empty interior. Here the bad locus has nonempty interior *and* swallows the target: the topological input of the continuation machine does not merely fail to be certified, it is false. Figure 1 shows the geometry.

Remark 4.4 (The refuted hope). The branch cut of the holomorphic Mahler measure $\tilde{m}$ (via Rogers' ${}_5F_4$, with $c^4 \in (0, 256]$) is the cross $[-4, 4] \cup i[-4, 4]$, one-dimensional. Had the identity $m(P+c) = \operatorname{Re} \tilde{m}(c)$ held inside $\mathcal{D}$ off the cross (the Villegas criterion [6]), then $c = 3$ would have been a boundary point of the good set and the differential-comparison machine of [1] would have ported verbatim. The go/no-go experiment `diag_n4_astroid.py` sampled τ with $c(\tau)$ inside the astroid *and off the cross* and compared $EK_4(\tau)$ against direct integration: the identity fails there too (differences 0.35–5.15). The bad set is not the cross; see the next section.

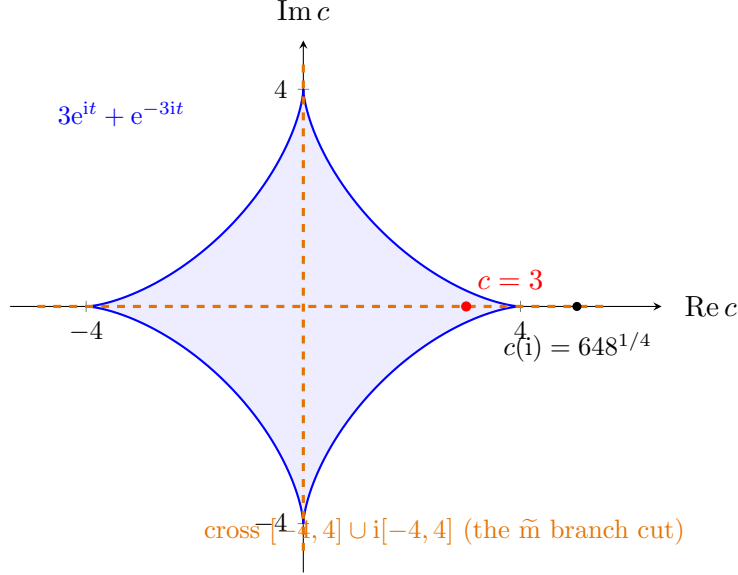


Figure 1: The critical image $P(\mathbb{T}^3)$ of the n_4 -family: the astroid disc $\mathcal{D}$ of (19) (shaded), bounded by $c = 3e^{it} + e^{-3it}$. The parameter $c = 3$ of the conjecture $n_4(81)$ is an interior point, so no holomorphic path in τ -space can reach τ_2 through the good region (Proposition 4.2). For comparison, the proved anchor $c(i) = 648^{1/4} \approx 5.04$ lies outside. The dashed cross is the one-dimensional branch cut $c^4 \in (0, 256]$ of the Rogers ${}_5F_4$ branch of $\tilde{m}$ — the bad set the $k = 1$ machine of [1] would have had to avoid, had $m = \text{Re } \tilde{m}$ held inside $\mathcal{D}$ off the cross; Section 5 shows it does not.

5 Obstruction 2: the U -series leaves the geometric sheet below $\text{Im } \tau = 1/\sqrt{2}$

The decisive obstruction is analytic rather than topological. Samart’s identity (10) is proved only for $\text{Im } \tau \geq 1/\sqrt{2}$, while

$$\text{Im } \tau_2 = \frac{\sqrt{7}}{4} = 0.66143\dots < \frac{1}{\sqrt{2}} = 0.70710\dots$$

The U -series (8) nevertheless converges for every $\tau \in \mathbb{H}$, defining EK_4 as a single-valued real-analytic function on all of $\mathbb{H}$; the function $\tau \mapsto n_4(s_4(\tau))$, by contrast, is real-analytic only off the preimage of the critical image. The question is where the two agree.

The diagnostic experiment of the project archive (`diag_n4_astroid.py`, 2026-07-29; control checks first: $\text{EK}_4(i) = 4m(P + 648^{1/4}) = 6.4332830658\dots$ to 5.4×10^{-12} , and conjugate symmetry exact) gives a clean verdict:

- *Below the threshold, the identity fails everywhere*, not merely inside the astroid: at $\tau = 0.3i$ one has $c(\tau) = 13.72\dots$ (real, outside $\mathcal{D}$, where even Rogers’ series is valid) and $|\text{EK}_4(\tau) - 4m(P + c(\tau))| = 11.4\dots$; at interior non-cross sample points the differences range over 0.35–5.15, with EK_4 taking *negative* values (while a Mahler measure is nonnegative).
- *The difference grows continuously from the boundary*: along the vertical scan $\text{Re } \tau = 1/4$, $\text{Im } \tau$ from 0.70 down to 0.20 (20 points), $|\text{EK}_4 - 4m|$ grows continuously from 0 at the threshold — no jump, hence no locally constant sheet offset that a correction term could fix.

Figure 2 shows the same phenomenon on the imaginary axis, computed afresh for this note (`gen_fig_n4.py`, `n4_departure.dat`): the two curves agree for $y \geq 1/\sqrt{2}$ (as they must by Theorem 2.2) and separate strictly below it, even though $c(iy)$ remains real and ≥ 4 — in particular outside the critical image — on the whole axis.

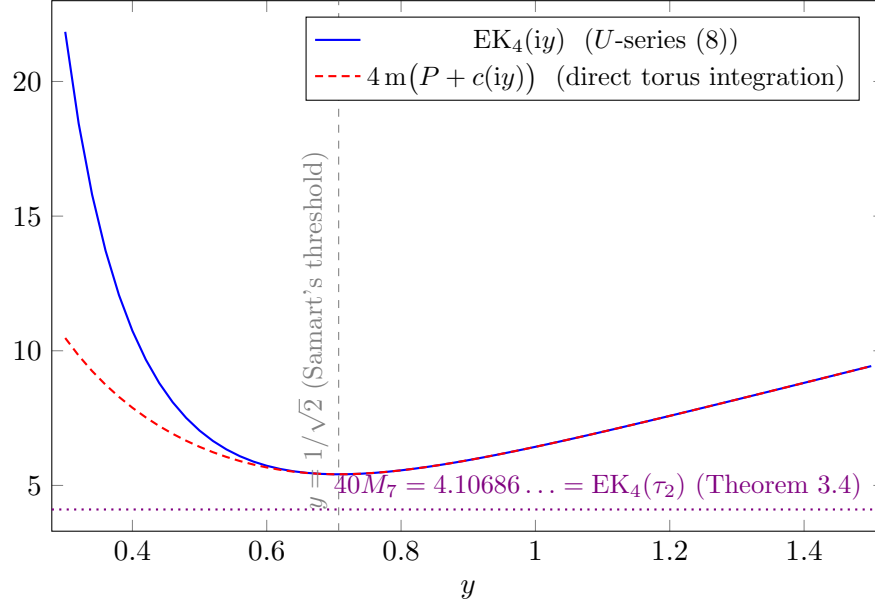


Figure 2: The wrong-sheet departure on the imaginary axis (data: `n4_departure.dat`, generated by `gen_fig_n4.py`; 61 ordinates, 30 dps, control $EK_4(i) = 4m = 6.4332830658\dots$). The U -series $EK_4(iy)$ coincides with the true Mahler value $4m(P + c(iy))$ for $y \geq 1/\sqrt{2}$ and departs from it below the threshold, although $c(iy) \geq 4$ stays real and outside the critical image on the whole range: the series has climbed to a companion sheet of $\tilde{m}$. The horizontal dotted line marks $40M_7 = 4.10686431115608\dots$, the exact value of the series at the CM point $\tau_2 = (7 + \sqrt{-7})/4$ (Theorem 3.4), i.e. the conjectured value of $n_4(81)$; it lies below every value the series attains on the imaginary axis. The direct integration of Section 6 shows $n_4(81) \neq EK_4(\tau_2)$: the series value is on the wrong sheet.

Remark 5.1 (Mechanism, and why the $k = 1$ machine does not port). This is the same wrong-sheet phenomenon as at the point τ' of Theorem B in [1] (Remark (wrong-sheet) there): EK is single-valued on $\mathbb{H}$, and below Samart's threshold it continues on a companion sheet of the multivalued $\tilde{m}$, with an exact closed-form value on that sheet (for n_2 : $(8/7)(44M_7 - d_7)$ instead of $n_2(s_2(\tau'))$). For Theorem A the continuation machine still worked because on the good component W the series branch *equals* the geometric branch: the differential identity $\Psi = \Omega(c)c' + E' \equiv 0$ could spread from Samart's region by the identity theorem, and the bad slit was avoidable. For the n_4 -family both premises fail simultaneously: the series branch differs from the geometric branch everywhere below $1/\sqrt{2}$ (so $\Psi \equiv 0$ is false on any would-be W), and the critical image is a disc around the target (Proposition 4.2). The Fricke shortcut of Theorem B is likewise unavailable: EK_4 is 1-periodic and the evaluation at $\tau_2 = \tau_w + 2$ is on the wrong sheet already. What remains is regulator/monodromy machinery in the style of He–Ye [7] (who proved Samart's f_3 conjectures), or a direct regulator evaluation at $c = 3$ in the style of Zheng–Guo–Qin [8].

6 Numerical status of the conjecture

A high-precision direct torus integration of

$$n_4(81) = 4 \operatorname{m}(x^4 + y^4 + z^4 + 1 + 3xyz)$$

gives

$$n_4(81) = 4.1655349907533676508(5),$$

against the conjectured value

$$40M_7 = 4.1068643111560804484182637959317165599558 \dots$$

(computed from $L(g_7, 3)$ via the functional equation (4)). The discrepancy is

$$n_4(81) - 40M_7 = +0.0586706795972872024 \dots,$$

about 2×10^5 times the integrator’s error floor 2×10^{-20} (outer tanh–sinh quadrature at degrees 5, 6, 7, 8, successive variations 6×10^{-21} , 1.9×10^{-20} , 4×10^{-21} ; cross-validated against an independent numpy trapezoid computation to 2×10^{-8}). Since $\operatorname{Im} \tau_2 = \sqrt{7}/4 < 1/\sqrt{2}$, this is exactly the regime in which the U -series is on the wrong sheet (Section 5), and the numerical picture is consistent: the series value $\operatorname{EK}_4(\tau_2) = 40M_7$ is exact (Theorem 3.4) but is *not* the true Mahler measure.

As literally stated, the conjecture $n_4(81) \stackrel{?}{=} 40M_7$ is therefore numerically refuted. What remains open is a closed form for the true value $n_4(81)$; the offset $0.0586706795972872 \dots$ has no identified closed form. We stress that this computation is high-precision numerical evidence, not an interval-arithmetic certificate in the sense of [1, §6]; with a discrepancy five orders of magnitude above the error floor we regard the conclusion as decisive.

The original evidence. Samart’s convention throughout [3] is that “ $\stackrel{?}{=}$ ” denotes equality to at least 25 decimal places, but no decimal value of $n_4(81)$ is printed there. In view of Fei’s caveat cited in §1 and of the exact series evaluation $\operatorname{EK}_4(\tau_2) = 40M_7$ (Theorem 3.4), the original numerical evidence was presumably series-sided. The same phenomenon in a two-variable family — failure of such identities inside the critical region “because its Deninger path is not closed” — has been studied explicitly by Samart and Tao [9]; the present example appears to be the first three-variable instance.

The integrator. The integrand is log-singular ($c = 3$ lies inside the critical image, §4). After the Jensen reduction in z , the inner integrand $g(t_1, t_2) = \sum_j \log^+ |z_j|$ has period $\pi/2$ in t_1 and symmetry about $\pi/4$, so $\operatorname{m} = (4/\pi^2) \int_0^{\pi/4} (\int g dt_2) dt_1$. The kink locus $\{|z_j| = 1\}$ of the z -Jensen reduction is detected *exactly*: the resultant of $z^4 + Az + B$ against its conjugate reciprocal is a degree-8 polynomial in $W = y^4$, $X = x^4$ (generated once by `kink_resultant.W.py`), whose unit-circle roots locate every kink; only two critical t_1 occur in $(0, \pi/4)$, namely $T_{1A} = 0.361367123906707805589029 \dots$ (where two kink pairs merge simultaneously) and $T_{1B} = \pi/6$. The inner integral is evaluated by tanh–sinh quadrature at degree 7 (degree 7 vs 8: 1.3×10^{-25}). The computation is implemented in `n4_81.final.py` (42-digit arithmetic).

7 Conclusion

As literally stated, the conjecture $n_4(81) = 40M_7$ is numerically refuted (Section 6): the two sides differ by $+0.0586706795972872\dots$, five orders of magnitude above the integration error floor. This note has archived its status:

- the L -value half is proved exactly: $EK_4(\tau_2) = 40M_7$ (Theorem 3.4), with complete algebra ($\Lambda_2 = \mathcal{O}_K$, $\Lambda_1 = (\bar{\pi})/2$, $-T_1 + 4T_2 = 28L(g_7, 3)$, exact $\zeta_K(2)$ -cancellation) and machine certification (45 checks, worst 2.7×10^{-59});
- the Mahler half is blocked twice: the critical image is the two-dimensional astroid disc with $c = 3$ interior (Propositions 4.1 and 4.2, Figure 1), and the U -series is on the wrong sheet of $\tilde{m}$ everywhere below $\text{Im } \tau = 1/\sqrt{2}$ (Figure 2);
- the numerical comparison of the two sides at $s = 81$ (Section 6) exhibits the wrong-sheet offset directly: $n_4(81) - 40M_7 = +0.0586706795972872\dots$, with integration error $\leq 2 \times 10^{-20}$.

The L -value half of the conjecture is thus proved exactly, while the literal identity fails: $EK_4(\tau_2) = 40M_7$ is the value of the U -series on the wrong sheet of $\tilde{m}$, not the true Mahler measure. The open problem that remains is a closed form for the true value $n_4(81)$, for which the methods of the companion paper [1] are structurally insufficient and regulator/monodromy techniques [7, 8] appear necessary.

Script inventory. `verify_P1_n4.81.py` (P1 certification, 45 checks); `diag_n4_astroid.py` (go/no-go diagnostic, wrong-sheet quantitative data); `gen_fig_n4.py` + `n4_departure.dat` (Figure 2, this paper); `kink_resultant.W.py` + `n4.81.final.py` (exact kink detection and direct integration at $c = 3$, Section 6). The main-paper scripts certifying the shared ingredients ((3), (4)) are listed in [1, §6]. All scripts of this note, together with their run logs, are available in the public repository <https://github.com/huiminZheng-collab/samart-mahler>, archived at <https://doi.org/10.5281/zenodo.21711885>.

References

- [1] H. Zheng, *Mahler measures at interior CM points: proofs of two conjectures of Samart*, companion paper (this repository, `paper/formal.tex`), 2026.
- [2] D. Samart, *Three-variable Mahler measures and special values of modular and Dirichlet L -series*, Ramanujan J. **32** (2013), no. 2, 245–268. arXiv:1205.4803.
- [3] D. Samart, *Mahler measures as linear combinations of L -values of multiple modular forms*, Canad. J. Math. **67** (2015), no. 2, 424–449. arXiv:1303.6376.
- [4] J. Fei, *Mahler measure of 3D Landau–Ginzburg potentials*, Forum Math. **33** (2021), no. 5, 1369–1401. arXiv:2009.09701.
- [5] M. Rogers, *Hypergeometric formulas for lattice sums and Mahler measures*, Int. Math. Res. Not. IMRN (2011), no. 17, 4027–4058.
- [6] F. Rodriguez Villegas, *Modular Mahler measures, I*, in: Topics in Number Theory (University Park, PA, 1997), Math. Appl. **467**, Kluwer, Dordrecht, 1999, 17–48.

- [7] Q. He and D. Ye, *On conjectures of Samart*, Manuscripta Math. **167** (2022), no. 3–4, 545–588.
- [8] H. Zheng, X. Guo, and H. Qin, *The Mahler measure of $(x + 1/x)(y + 1/y)(z + 1/z) + \sqrt{k}$* , Electron. Res. Arch. **28** (2020), no. 1, 103–125.
- [9] D. Samart and Z. Tao, *Determinants of Mahler measures and special values of L -functions*, arXiv:2409.14714.
- [10] The LMFDB Collaboration, *The L -functions and Modular Forms Database*, newform 7.3.b.a, <https://www.lmfdb.org>.
- [11] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic* (version 1.4.1), <https://mpmath.org>.